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TypeScript-first schema validation with static type inference  
by [@colinhacks](#)

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Zod 4 is stable. Read the [release notes](#) and [migration guide](#).

# Introduction

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Zod is a TypeScript-first validation library. Using Zod, you can define *schemas* you can use to validate data, from a simple `string` to a complex nested object.

```
import * as z from "zod";

const User = z.object({
  name: z.string(),
});

// some untrusted data...
const input = { /* stuff */ };

// the parsed result is validated and type safe!
const data = User.parse(input);

// so you can use it with confidence :)
console.log(data.name);
```

## Features

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- Zero external dependencies
- Works in Node.js and all modern browsers
- Tiny: 2kb core bundle (gzipped)
- Immutable API: methods return a new instance
- Concise interface
- Works with TypeScript and plain JS
- Built-in JSON Schema conversion
- Extensive ecosystem

# Installation

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```
npm install zod
```

 Zod is also available as `@zod/zod` on [jsr.io](https://jsr.io).

Zod provides an MCP server that can be used by agents to search Zod's docs. To add to your editor, follow [these instructions](#). Zod also provides an [llms.txt](#) file.

## Requirements

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Zod is tested against *TypeScript* v5.5 and later. Older versions may work but are not officially supported.

### "strict"

You must enable `strict` mode in your `tsconfig.json`. This is a best practice for all TypeScript projects.

```
// tsconfig.json
{
  // ...
  "compilerOptions": {
    // ...
    "strict": true
  }
}
```

## Ecosystem

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Zod has a thriving ecosystem of libraries, tools, and integrations. Refer to the [Ecosystem page](#) for a complete list of libraries that support Zod or are built on top of it.

- [Resources](#)
- [API Libraries](#)
- [Form Integrations](#)
- [Zod to X](#)
- [X to Zod](#)
- [Mocking Libraries](#)
- [Powered by Zod](#)

I also contribute to the following projects, which I'd like to highlight:

- [tRPC](#) - End-to-end typesafe APIs, with support for Zod schemas
- [React Hook Form](#) - Hook-based form validation with a [Zod resolver](#)
- [zshy](#) - Originally created as Zod's internal build tool. Bundler-free, batteries-included build tool for TypeScript libraries. Powered by `tsc`.

## Sponsors

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Sponsorship at any level is appreciated and encouraged. If you built a paid product using Zod, consider one of the [corporate tiers](#).

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### Gold

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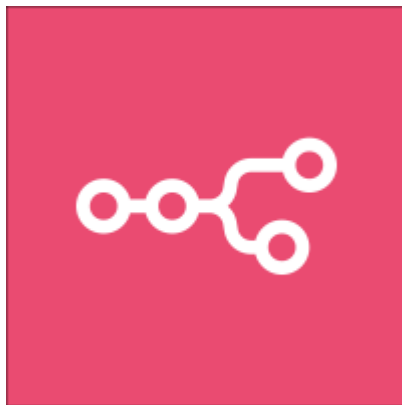


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## Bronze



**Basic usage >**

Basic usage guide covering schema definition, parsing data, error handling, type inference, an...